

$$\begin{bmatrix} 2 & 3 & 5 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 11 \\ -3 \end{bmatrix}$$

①

$$3x_3 = -3 \rightarrow \underline{x_3 = -1}$$

$$4x_2 + x_3 = 11 \rightarrow 4x_2 = 12 \rightarrow \underline{x_2 = 3}$$

$$2x_1 + 3x_2 + 5x_3 = 0$$

$$2x_1 + 9 - 5 = 0 \rightarrow 2x_1 = -4 \rightarrow \underline{x_1 = -2}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

②

PDR #2

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$$f(x) = 2x^5 - 13x^4 + 26x^3 - 32x^2 + 28x - 10 \quad (3)$$

UB due to Cauchy:

$$1 + |a_n|^{-1} \cdot \max_{0 \leq i \leq n-1} |a_i| = 1 + \frac{1}{2} \cdot 32 = 17 = \underline{P_2}$$

$$LB: S(x) = x^n f(x^{-1}) = -10x^5 + 28x^4 - 32x^3 + 26x^2 - 13x + 2$$

$$1 + |a_n|^{-1} \cdot \max_{0 \leq i \leq n-1} |a_i| = 1 + \frac{1}{10} \cdot 32 = 4.2 = P_1^{-1}$$

$$\underline{\frac{5}{21} = P_1}$$

$$P(x) = 3x^5 - 32x^4 + 140x^3 - 315x^2 + 365x - 174$$

$$x_0 = 2$$

4

$$\begin{array}{l}
 2 \left| \begin{array}{cccccc} 3 & -32 & 140 & -315 & 365 & -174 \\ \downarrow & 6 & -52 & 176 & -278 & 174 \end{array} \\
 2 \left| \begin{array}{cccccc} 3 & -26 & 88 & -139 & 87 & 0 \\ \downarrow & 6 & -40 & 96 & -86 & \end{array} \rightarrow \frac{P(2)}{0!} \\
 2 \left| \begin{array}{ccccc} 3 & -20 & 48 & -43 & 1 \\ \downarrow & 6 & -28 & 40 & \end{array} \rightarrow \frac{P'(2)}{1!} \\
 2 \left| \begin{array}{cccc} 3 & -14 & 20 & -3 \\ \downarrow & 6 & -16 & \end{array} \rightarrow \frac{P''(2)}{2!} \\
 2 \left| \begin{array}{ccc} 3 & -8 & 4 \\ \downarrow & 6 & \end{array} \rightarrow \frac{P'''(2)}{3!} \\
 2 \left| \begin{array}{cc} 3 & -2 \\ \downarrow & \end{array} \rightarrow \frac{P^{(4)}(2)}{4!} \\
 3 \rightarrow \frac{P^{(5)}(2)}{5!}
 \end{array}$$

Taylor Polynomial:

$$P_n(x) = \sum_{i=0}^n \frac{P^{(i)}(x_0)}{i!} (x-x_0)^i$$

$$x_0 = 2$$

$$P_5(x) = \sum_{i=0}^5 \frac{P^{(i)}(x_0)}{i!} (x-x_0)^i =$$

$$0(x-2)^0 + 1(x-2)^1 + (-3)(x-2)^2 + 4(x-2)^3 + (-2)(x-2)^4 + 3(x-2)^5$$

TH

